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on Data Science, Machine Learning and It's Application.

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9th and 10th October, 2020

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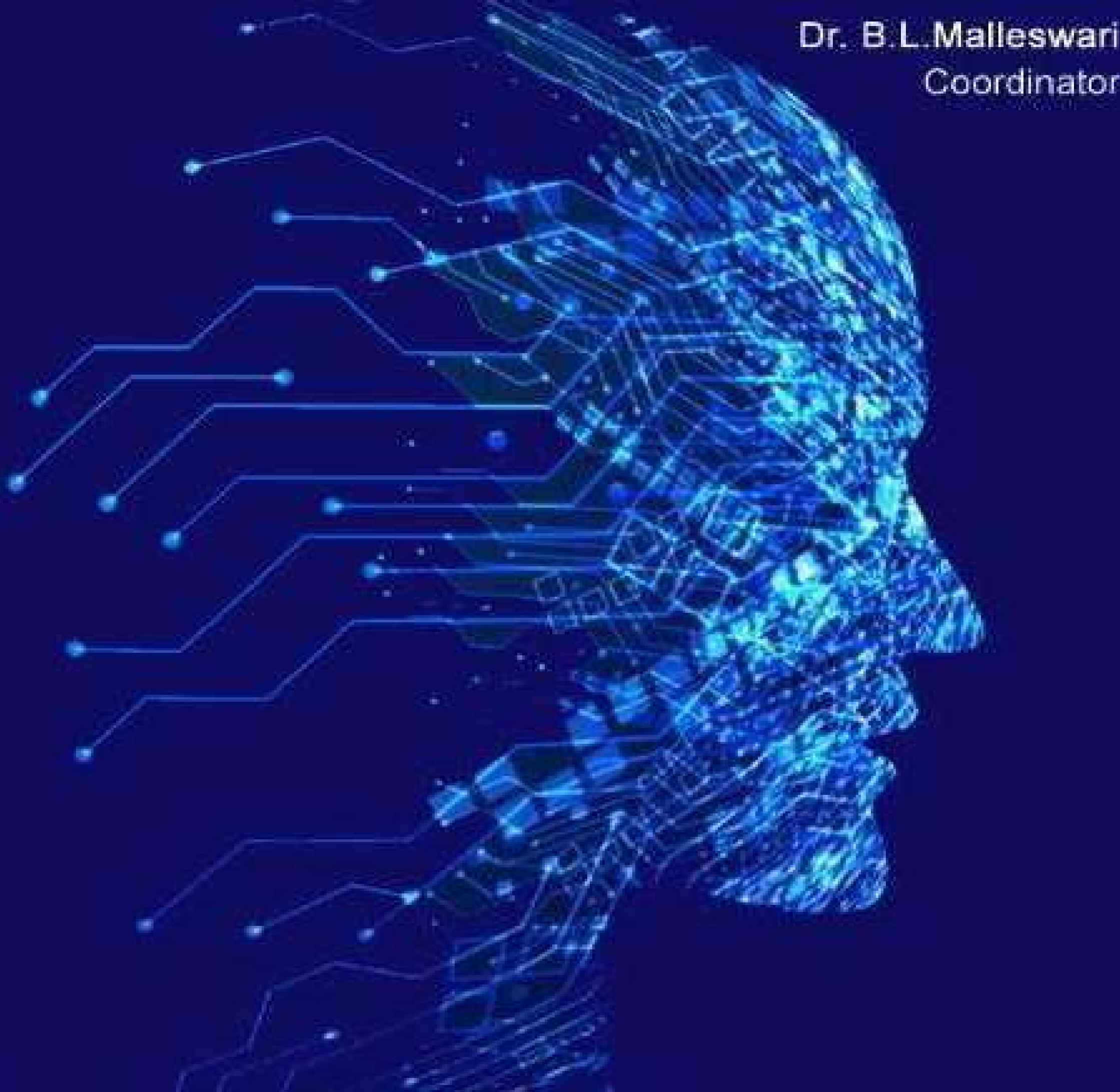

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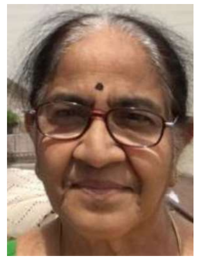


It gives me immense pleasure to be a part of the Two day AICTE sponsored Online International Conference on “Data Science, Machine Learning and its applications”. I would like to congratulate the entire SWEC team for organizing this conference and wish you all success in its execution.

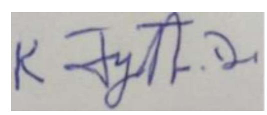
ICDML2020 is a perfect platform to collaborate with experts into Data Science and Machine Learning who will showcase high-end research papers and its applications that can solve social and economic problems.

This pandemic has changed the way we use technology and stay remotely connected and yet be productive. Smart cities, digital learning and work from home have become the “new normal”.

ICDML2020 aims to witness knowledge sharing and showcasing of machine learning and data science products by scientists and practitioners from academia and Industry. Be it any industry today; Information Technology, Banking, Finance and Healthcare automation and innovation is revolutionizing almost everything.

This conference will discuss and bring everything which may be of interest to persons with an inclination towards computing, Data Science and Machine Learning.

All the best to you all!



Smt. K. Jyothi Devi
Chairperson
Sridevi Education Society

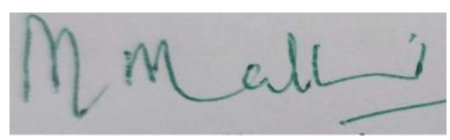
Prinicipal Message

It is my privilege to announce that the proceedings of ICDML2020 are ready for publication. The conference proceedings give an insight into the scope of the imagination, creativity and research depth of our participants. I applaud the editorial team for the hard work and dedication they have invested in realizing this goal, and wish all the participants and colleagues great success in all their future endeavours.

ICDML2020 conference aims to provide a platform for researchers across the globe to present their innovative ideas and share technical knowledge on Data Science, Machine Learning and their applications. This conference encourages own copyright papers in the diverse domains pertaining to the field of Data Science and Machine Learning. The participants will be benefited by mutual exchange ofideas and research innovations. The conference proceedings aim to publish quality and unpublished work.

I congratulate the team of faculty for their tireless efforts that have come to fruition in the form of this conference proceeding. I wish it all success and hope that this tradition that has been set by the current team will be carried through by the following generation of the college.

I express my sincere gratitude to the Management of SWEC for motivating and supporting us in the path to realize our vision and enable us to take the Institute to scale new heights.



Dr. B.L. Malleswari
Principal
Sridevi Women's Engineering College

Editorial

We are delighted to bring out the ICDML2020 proceedings despite the current pandemic situation. It is a moment of pride for us to have organized the Two day AICTE sponsored Online International Conference in Data Science, Machine Learning and its applications. The aim of the conference is to publish good quality papers that are original and credible. The papers of the scholars have significantly contributed to the effectiveness and equitability of scientific information disseminated.

The participants selected their topics from a wide range of options and published their research papers that demonstrate knowledge and expertise.We are glad to have received good feedback from the multidisciplinary audience who participated in the conference.

We received close to321 abstracts from research scholars, faculty and professors. The technical committee reviewed and selected the best papers from those for oral presentations. 70 papers were selected and they were given the opportunity to present. This volume contains 36 Data Science and 34 Machine Learning papers. Research papers published in this issue are expected to be of much interest to the readers.

On behalf of all the committees, I would like to express my deep gratitude to everyone who has showed immense interest and passion to submit such a variety of abstracts, share their presentations and articles for publication.

I would like to thank the Technical Committee and the Advisory Board for their expertise and support in reviewing and shortlisting the research papers.

I would like to thank the Editorial committee who in coordination with other committees has showed us an exemplary example of team work and excellent planning in bringing out this issue on time.

Dr. S Jagadeesh
Editor
SWEC

Message from Guest

Smt. P. SABITHA INDRA REDDY
MINISTER FOR EDUCATION
Government of Telangana.

I am glad to note that Sridevi Women's Engineering College is organizing a two day online International Conference and would like to congratulate the Management, the Principal and the faculty of Sridevi Women's Engineering College.

I am sure this conference on Data Science, Machine Learning and its applications will offer a great platform to the Academicians, Research Scholars, Faculty and Industry practitioners for the exchange of the latest developments, ideas and research.

I wish this conference to be a splendid event, both in terms of intellectual quality and social gratification. You have the right mix of topics, diverse set of speakers and high-level array of participants that will definitely make it a successful event.

My best wishes to you all for making it very engaging and insightful conference with great learning experience during this pandemic situation.

(P.SABITHA INDRA REDDY)

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ABOUT ICDML 2020

Sridevi Women's Engineering College is organizing an International Conference sponsored by AICTE on Data Science, Machine Learning and its Applications (ICDML2020).

The conference provides a premier forum that brings together Research Scholars, Faculties, Industry practitioners, as well as developers and users in Electronics, Electrical, Computer Science, and Information technology for the exchange of the latest theoretical developments in Data Science and Machine Learning and their applications.

The Conference invites submission of papers describing innovative research on all aspects of Data Science and Machine Learning as well as application-oriented papers that make significant, original, and reproducible contributions to improving the practice of Data Science and Machine Learning analytics in real-world scenarios.

ICDML2020 offers a knowledge sharing opportunity at two levels – research track and use of new research methodology.

ICDML2020 will also feature a peer-reviewed Industry session, whose purpose is to showcase recent and early-stage research developments on topics that are of interest to Faculty and Industry practitioners in Data Science and Machine Learning.

CONFERENCE TOPICS

We welcome original research papers on all aspects of data science in relation to machine learning, including but not limited to the following topics:

* Auto-ML
* Fusion of information from disparate sources
* Feature engineering, Feature embedding, and data
* preprocessing
* Learning from network data
* Learning from data with domain knowledge
* Evaluation of Data Science systems
* Risk analysis
* Causality, learning casual models
* Deep learning
* Autonomous systems
* Analysis of Evolving Social Networks
* Embedding methods for Graph Mining
* Online Recommender Systems
* Augmented Reality, Computer Vision
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ABOUT SWEC

Sridevi Women's Engineering College (SWEC) was established in the year 2001 with the approval from AICTE, New Delhi and Government of A.P. The College is affiliated to J.N.T.U. Hyderabad and is sponsored by Sridevi Education Society. The core objective of the institution is to contribute to the Educational needs of State and Nation, with an emphasis on promoting girl education and women empowerment.

Sridevi Women's Engineering College provides an excellent academic environment and latest learning resources to help students make the most out of their career goals.

The institute, besides aiming at shaping its students into healthy and disciplined young citizens of character and culture, places emphasis on practical and professional experience so as to enable them to secure employment in industries immediately after completion of their studies and also become entrepreneurs. The Institute is developing close contact with a large number of industries situated in and around Hyderabad for mutual benefit.

Vision
To attract the finest talent by creating an atmosphere conducive to learning and to train and empower female professionals with global skills thereby assigning their legitimate place of honour in the society.

Mission
To create a center for excellence in engineering by imparting knowledge and skills.
To facilitate women empowerment by enhancing their technical competency and intellectual capabilities.
To create a conducive work environment with an equally competent and experienced team that derives strength from each other.

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Papers are invited on below listed Topics:

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* Fusion of information from disparate sources
* Cyber security and Forensic science
* Learning from network data
* Learning from data with domain knowledge
* Evaluation of Data Science systems
* Risk analysis
* Causality, learning casual models
* Deep learning
* Autonomous systems
* Analysis of Evolving Social Networks
* Embedding methods for Graph Mining
* Online Recommender Systems
* Augmented Reality, Computer Vision
* Green Data sciences
* IoT data analytics and Big Data
* Cloud computing

Contributions must contain new, unpublished, original and fundamental work relating to the Machine Learning journal's mission. All submissions will be reviewed using rigorous scientific criteria whereby the novelty of the contribution will be crucial.

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Deep Learning and Its Applications: A Review

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Abstract

Deep learning is a new field of machine learning (ML). It requires a variety of layers of synthetic neural networks. Deep knowledge of the method includes nonlinear shifts and irregular state process discussions in large datasets. Ongoing developments in deep learning systems in different fields have only made major contributions to human-made intellectual capacity. The whole paper provides the best description of his class on the achievements and innovative applications of deep learning. The analysis demonstrates chronologically how and in what practical implementations deep efficiency measurements have been used. In addition, the predominant and practical calculations of the theory of deep learning and its advancement in non-linear layers and behaviors are mirrored in more conventional calculations in simple applications. The best research in its class presents a general overview of the novel's ideas and desires and the ubiquity of ever-expanding profound thought.

Keywords--Deep learning, Machine Learning, Applied Deep Learning.

I. Introduction

Man-made Artificial Intelligent Ability (AI) as a machine-shown view has become an important way of approaching human learning and thinking [1]. In 1950 "The Turing Test" was suggested as a fascinating example of how a PC could evolve reasoning about human opinion [2]. AI is segregated in increasingly specific subfields of science as a subject of study. For example: natural language processing (NLP)[3] can boost the compositional participation in various applications[4][17]. Mama Chine's interpretation calculations have produced a range of applications which consider the structure of grammar harm to be spelling errors. Furthermore, as the PC suggests improvements to the author or boss, often phrases and terms associated with the fundamental concept are usually used as the primary source [5]. Figure 1 describes in depth how the AI encompasses seven Computing research subfields. Machine learning and the retrieval of knowledge have lately become the focal point of concern and the most relevant problems within the research network. There are Mixed Research Areas exploring many potential consequences in serving a database[9]. By the way, in the last decades, only excellent techniques and measurements have been used to process this material. While improving these calculations could lead to a feasible self-study[19]. One of the most critical aspects of this rationalization is prescribing, in which signs, conditions and rehabilitation strategies produce immense datasets that can be used to provide effective drugs[11]. Since ML covers a broad variety of study, various methodologies have been created. Bunching, the Bayesian Network, Deep Learning and Decision Tree Modeling are only a couple of the methodologies. The attached examination focuses primarily on deep assimilation, basic concepts, past and presents experiences in a number of fields.It also reveals some evidence showing the steady progress in deep learning study in the conceptual datasets, despite the obstacles of the current years.

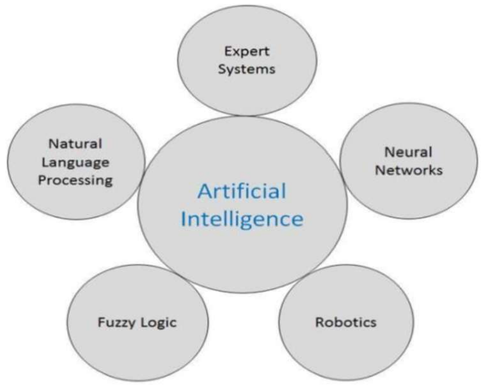

Figure 1. Research in artificial intelligence.

II. Background

Keep idea of Deep Learning (DL) appeared unprecedented for 2006 as another field of research in machine learning. In the first place it was also known as various levels of learning in [2]. It usually included many search fields identified with examples of recognition. In-depth recruitment mainly thinks of two key variables: nonlinear preparation in many layers or organizes and learns regulated or unsupervised [4]. The nonlinear preparation in several levels refers to a calculation in which the current level takes the performance of the last level as its performance. Between the levels a chain of value is formed to make up the meaning of the knowledge deemed important or not. On the other hand, controlled and unmonitored implementation is related to the product category of the class; its functionality indicated a significant structure, whereas lack of assistance indicates the unsupervised model.

III. Applications

Deep learning implies an analysis of the mental framework and techniques at many stages. In almost any event, it continues to be used in many original applications.. For example, in the preparation of computer images, shading the dark-scale image of an image was physically done by customers who had to choose each shade based on their own criteria. By applying an in-depth learning calculation, shading can be done naturally from a PC [10]. Additionally, the solid can be included in a silent battery video using recurrent neural networks (RNN) as an important aspect of deep learning techniques [18]. Deep learning can be understood as a technique to improve results and improve the time of professional things in some forms of calculation. In the field of the cessation of the normal dialectal star, deep learning strategies have been linked to the age of the image captions [20] and the age of calligraphy [6]. The accompanying applications are ordered in preparation of the image, prescription and unmodified biometrics. In the below fig 2 shows the different applications of machine learning

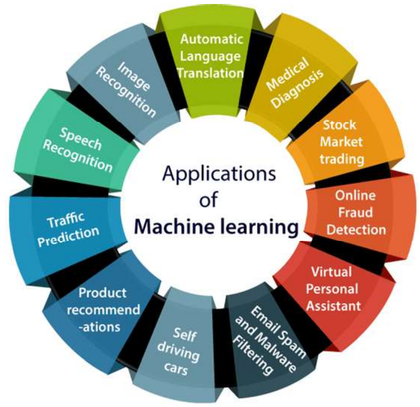

Figure 2. Machine learning applications.

Image Processing: It was produced by applying sieve molecules and proliferation of the Bayesian belief. The core premise of this program indicates that a human being can interpret an individual's material by viewing only a picture of the partly cut face[14], a PC may in this way replicate the picture of a transformed face. Later in 2006, the calculation of insatiability and the progressive system came together in an application capable of processing manually written figures [7]. Ongoing consultations have related the key method for advanced image processing to deep learning. For e.g., it might be more feasible to add a neural convolution network (CNN) to iris recognition than using traditional iris sensors. The reliability of CNN will hit a cumulative accuracy of 99.35 percent[16].The recognition of a versatile area these days allows the customer to achieve a specific position that depends on an image. Supervised semantics: the deep hash preservation algorithm (SSPDH) demonstrated significant improvements in the Visual Hash Bit (VHB) and Space-Saliency Fingerprint Collection (SSFS) studies. SSPDH accuracy level is even more powerful at 70% [15].Also, facial recognition is another impressive technology of advanced image processing, which uses a deep learning technique. Google, Facebook, and Microsoft have excellent examples of powerful face detection learning models [8].Recently, the recognizable test that depends on by defining age and sexual orientation as introductory criteria the facial expression has now become a programmed identification. Sighthound Ltd., for example, attempted a profound calculation of the convolutional neuronal system capable of perceiving age and sex, as well as feelings [3]. In addition, a robust system was developed to reliably assess the age and gender identity of a person from a solitary picture using a fundamental framework of multitasking learning[21].

Medicine: Computerized image processing is a vital aspect of the area of analysis where a deep learning approach can be applied. Medical systems have only been checked in this respect. For instance, analysis of shallow learning and deep learning in neural networks, led to superior implementation of disease expectations. An image taken of an MRI scan [22] of a human brain was prepared to anticipate a possible Alzheimer's disease [3]. While this process has been completed early, certain problems can be considered for potential implementations. High-caliber planning and reliance is a result of the constraints. Number, consistency and unpredictable nature of knowledge are research viewpoints, but heterogeneous union forms of information are a possible part of deep learning[17][23].Another example in which deep learning techniques suggest significant findings is the optical coherence tomography (OCT). Traditionally, photographs are viewed by manual enhancement of systems of transformation [12].Ironically, the lack of a fixed planning limits the methodology of deep learning. In any case, the introduction of improved preparedness packages within a couple of years will effectively avoid retinal pathologies and reduce the expense of OCT technology [24].

Biometrics: In 2009, the programme for speech recognition was planned to reduce the telecommunications error rate (PER) using two separate versions of deep conviction systems[18].In 2012, a hybrid neural network-Hidden Markov Model (NN – HMM)-connected the CNN technique[25]. As a result, a PER of 20.07 per cent was achieved. Acquired PER is better when tested with a neural system calibration technique recently connected to three layers[26].Iris identification has been tested on mobile phones and their photographic lenses. Using mobile phones developed by various organisations, the accuracy of iris detection will hit 87% of profitability[22][28]. “R. Vargas, A. Mosavi, L. Ruiz, Deep Learning: A Review, Advances in Intelligent Systems and Computing, (2017)”. As for security, in particular get to the control; Deep learning related to biometric attributes is used. DL has been used to accelerate the development and enhancement of guardian facial recognition technology. As suggested by this designer, their products could enhance the special testing phase from one to one in nine months [27]. Without a DL presentation, the motor development may have taken 10 men long. Speeding up the production and shipment of components. These devices are used at the Heathrow aircraft terminal in London and can possibly be used for the time and participation and in the management of an account segment [3] [29].

Learning Associates: Compared to broad databases, learning associates, also known as learning affiliate rules, are a technique for discovering interesting interactions between variables. Solid standards found in libraries that use certain interesting standard proportions are required to be remembered. Affiliate laws include the potential that / descriptions of assistance show links in a social network or other data warehouse for otherwise isolated records. One example of the Affiliate Guideline will be "If a consumer buys twelve eggs, it is 80 per cent likely to purchase drainage as well." Affiliation The law has two parts, a prerequisite (if any) and a conclusion. A predecessor has been identified in the details. A second is also something that can be found in conjunction with the precursor. Most machine learning algorithms used for data processing and information science work using numerical knowledge. In addition, in general, multiple equations would be extremely empirical. Digging the affiliate rules is suitable for transparent (non-numeric) knowledge and requires limited checking rather than strict checking.

Video Surviellance: Video monitoring entails the possibility of visual detection without being directly on-site.The observation can be performed consistently and can be recorded for a subsequent survey using information storage devices. Video recognition innovation could also be used for control and direction purposes such as traffic control, generation control and quality assurance. It offers several advantages, such as the recognition of crimes, the documentation of security-related occasions, and the action against vandalism, the monitoring of guilty people, the verification of outdoor land and the reduction of setbacks. North America dominates the general market for video recognition and all-inclusive VSaaS, contagious of the dangers of psychological oppression and the incessant mechanical progressions of security-related innovations in this place. Global video recognition and the VSaaS market will continue to show strong development due to growing concern for security worldwide. Advances in watching videos have proved incredibly vital, particularly in detecting the traffickers' exercises of fear and the likelihood of illegal invasions. Governments around the world are following these lines by putting resources into state-of-the-art Video surveillance equipment and advanced identification mechanisms to provide driving power for their strategic warfare operations. Bringing the baseline to video observation has become a general impetus, which makes market development useful.

F. Product Recommendations: Recommendatory programmers' are among the most popular and widely accepted deep learning technology implementations in enterprise. We may find large-scale suggestion networks for online, on-demand television or music streaming. In general, a firm wants a community of expensive data scientists and engineers to build and manage those programs. Similar machine learning algorithms may be used to monitor news or video reviews for the internet, product suggestions or optimization in travel and retail. Additionally these algorithms can be modified in each suggestion request using our special query language. In recommendation systems, machine learning algorithms typically fall into two categories: collective and content-based filtering strategies, while current recommendations incorporate both approaches. Content-based approaches are made using a combination of entity attributes and cooperation approaches are used to compute the correlation of encounters. In the occasional case where users have an clear evaluation of the components, communication methods function with the interaction matrix which can also be called the evaluation matrix. The goal of machine learning is to find a feature that offers the utility of the elements to each individual. Typically the matrix is large, very sparse and most values are absent. Laws of association can be derived using various algorithms. Recommendations may be tested in the same manner as traditional machine learning models in historical evidence (offline assessment).

IV. Overview

This table includes brief details on the new innovations, algorithms and scientists.

Table 1. Latest technologies with algorithms

Scientists	Latest technology	Algorithm
Richard Zhang , Phillip Isola	Automatic colorization of black and white image	Computer vision algorithm
Daniil Korbut	Automatic machine learning translation	Neural Network Technology
Paul Viola & Michal Jones	Object classification and detection in photographs	Faster Conventional Neural Network
Alex Graves	Automatic handwriting generation	Rapid Response Network Graphic Courtesy
C Shorten	Automatic Text generation	Rapid Response Network/Long Short Term Memory Sequential Generation
Daniel Ricciardelli	Automatic Caption Generation	Long Short Term Memory /Conventional Neural Network

V. Conclusion

Deep learning, also referred to as profound learning, is currently a fast-growing use of machine learning. The numerous implementations shown above indicate a dramatic change in just a few years. The use of these equations in diverse fields shows their adaptability. Obviously the representation made in this research It demonstrates the significance of this technique and gives a good explanation of the progress of deep learning and the potential for future research in this area. Also, remember that the order of layers and the control of learning are essential components in the development of a strong framework for deep learning. Hierarchical structure is necessary to modify information structures, while regulation recognizes the value of the records as part of the protocol. The fundamental calculation of deep learning relies on the success of current implementations of machine learning, owing to their creativity of handling various stages of learning. Deep learning can provide powerful results in advanced image management and speech recognition. Hence, deep learning is a contemporary and energizing phenomenon of the development of man-made intellectual capability, and has shown substantial change.

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